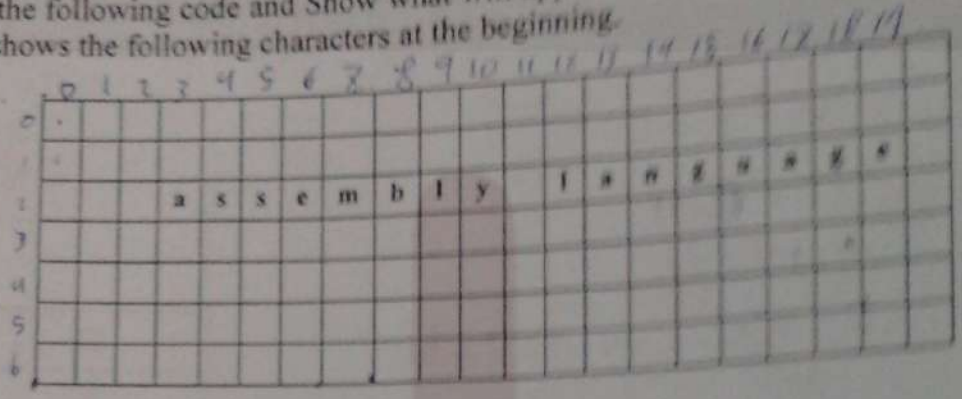
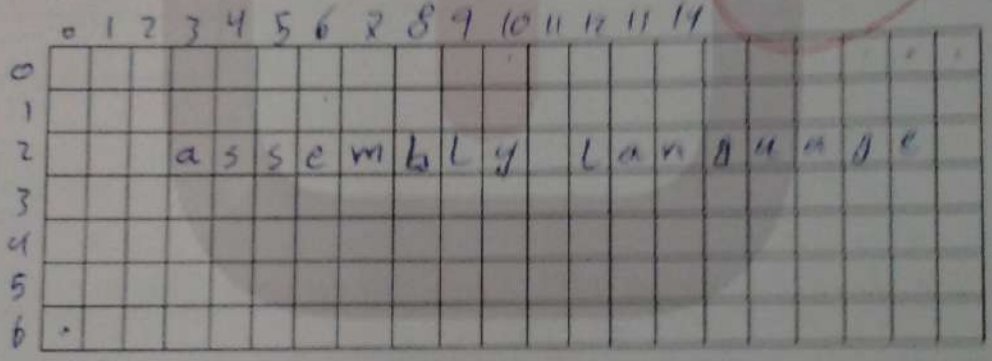
Q4) Trace the following code and Show what will appear on the screen after execution. Assume the screen shows the following characters at the beginning.



Draw the screen after executing the following code:

```
mov ah, 06h
mov al, 3
mov cx, 0006h
mov dx, 0709h
int 10h
```



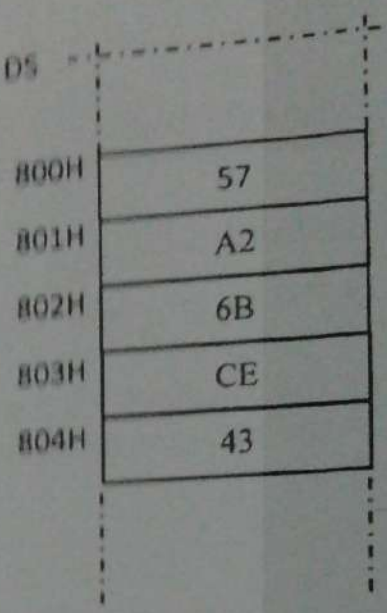
INT 10H		
AH	**Input Parameters**	**Description**
06	AL: Number of lines to scroll. BH: Screen attribute for cleared area. CL: x coordinate UL, CH: y coordinate UL DL: x coordinate LR, DH: y coordinate LR	Clear or scroll up.

GOOD LUCK

0111, 0101

Q1) Trace the following fragment of code and answer the questions below:

```
MOV AL, [800H]
MOV CL, 4
ROL AL, CL
MOV [800H], AL
```



A. What is the value of memory location (DS:800H) = ... 75 .....

B. Modify the previous code to apply the same operation on the 5 memory locations starting at (DS:800H). (Note: **use LOOP instruction**).

```
mov cx,5
mov si,800H
L1: push cx
mov AL,[si]
mov cl,4
ROL AL,cl
mov [si], AL
pop cx
inc si
Loop L1
```

Q2) Consider the following piece of stack memory, suppose that SS = 100h and that the physical addresses increase in the direction of the arrow. Trace the following code and fill the blanks with the correct answer:

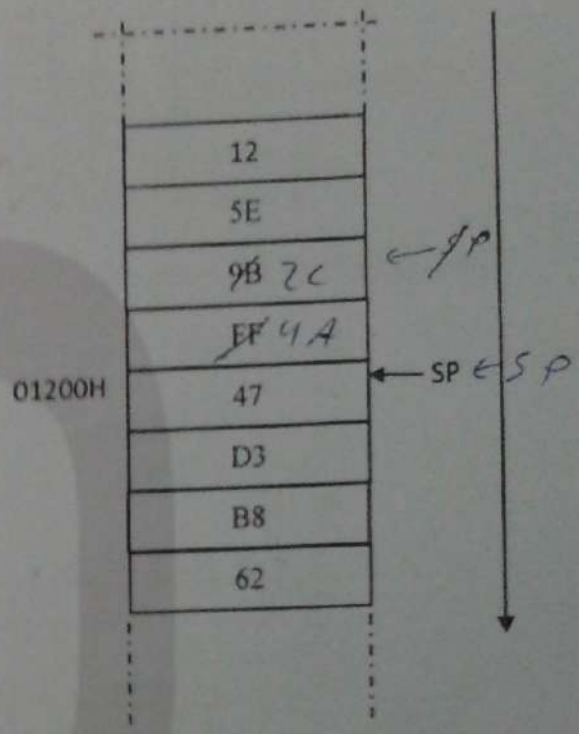
11FEH

PUSH 4A2CH ;SP= 11FEH

POP AX ;AX= 4A2CH

POP BX ;BX= D347, SP= 120219

3



Q3) Trace the following piece of codes and fill the blanks with the correct answers:

a)

```
MOV AL, -2
MOV BL, 2
MUL BL
```

Ax = 1E4H

b)

```
MOV AX, -5
MOV BL, 2
IDIV BL
```

Ax = FF02H

c)

```
X DB 'ABC'

MOV SI, 0
MOV CX, 3
L:  TEST X[SI], 01H
    JZ L1
    ADD X[SI], 20H
L1: INC SI
    LOOP L
```

X = 'aBc'

3

30/16

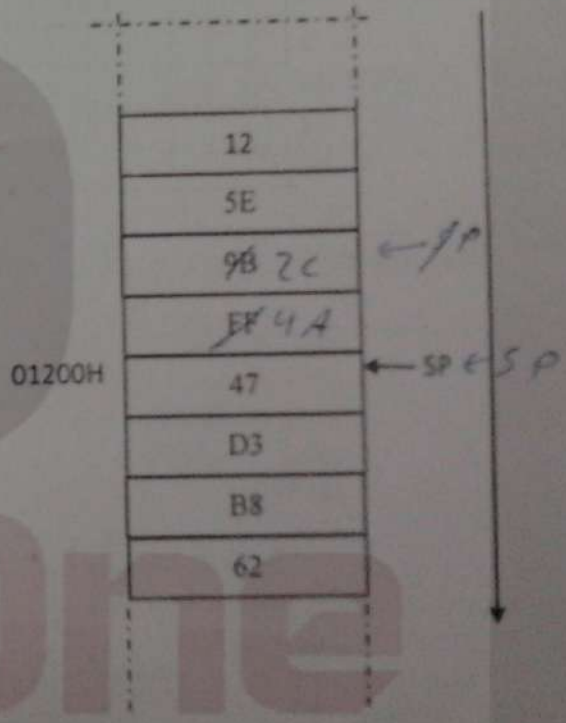
Q2) Consider the following piece of stack memory, suppose that SS = 100h and that the physical addresses increase in the direction of the arrow. Trace the following code and fill the blanks with the correct answer:

11FEH

PUSH 4A2CH ;SP= 11FEH

POP AX ;AX= 4A2CH

POP BX ;BX= D347, SP= 12021H



Zero One

...codes and fill the blanks with the correct answers: